Final Denominators in Modular Scattering Do Not Define Closed-Orbit Clocks: Stable Closure and a Route-A Obstruction

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Abstract

The lower-left denominator of a modular matrix carries genuine Riemann-zeta arithmetic: oriented cusp double cosets of denominator c occur with multiplicity $\varphi(c)$, and their Dirichlet series is the arithmetic part of the modular scattering coefficient $\Phi(s) = \Lambda(2s-1)/\Lambda(2s)$. We test whether this open-channel denominator can also define the total period of a closed hyperbolic orbit. For every fixed $\alpha > 0$, we prove that any function $F : \alpha\mathbb{N}_{>0} \to \mathbb{R}$ satisfying $F(\alpha|c(g^2)|) = 2F(\alpha|c(g)|)$ for every hyperbolic $g \in \mathrm{SL}_2(\mathbb{Z})$ whose four entries are strictly positive is identically zero. Thus no nontrivial clock depending only on the final monodromy denominator supports the standard primitive/repetition law. The obstruction is sharp: Cayley–Hamilton yields an exact defect formula and

$$\lim_{n\to\infty} \frac{2\log(\alpha|c(g^n)|)}{n} = \ell(g),$$

where $\ell(g)$ is the Selberg hyperbolic translation length. Hence the canonical stable closure returns to the known Selberg–Mayer clock. Exact integer certificates and an independent 110-digit checker audit 274 primitive even Gauss words; 259 have cyclic denominator variation and none has literal denominator square-additivity. The result closes one tempting dynamical-zeta route but does not exclude local, endpoint-extended, cohomological, or multi-cusp constructions.

1 Introduction

The modular scattering matrix contains the Riemann zeta function for a geometric reason, not because a target zero list has been fitted. For the one-cusp modular surface $\mathrm{PSL}_2(\mathbb{Z})\backslash\mathbb{H}$, its scalar coefficient is

$$\Phi(s) = \frac{\Lambda(2s-1)}{\Lambda(2s)}.$$

This makes modular scattering a natural high-risk candidate in a Hilbert–Pólya structure search. It also creates a categorical trap. The denominator that indexes an open cusp-to-cusp channel belongs to a double coset, while a primitive closed geodesic belongs to a hyperbolic conjugacy class. A dynamical Euler product needs the latter object and its exact repetition law.

The distinction matters for attempts to move beyond an area-preserving Hénon model of Riemann-zero statistics [9]. The local Hénon program motivates the search for a source-derived clock, but it does not prove that a fitted or continuum-regularized clock is intrinsic. We therefore change the system rather than assume the proposed bridge. Modular scattering supplies an exact arithmetic clock and lets us ask whether it survives closure.

We study the narrow proposal

$$R_F(g) = F(\alpha|c(g)|), \qquad \alpha > 0,$$

where $c(g)$ is the lower-left entry of a $\mathrm{SL}_2(\mathbb{Z})$ lift and R_F is meant to be the total period of a closed hyperbolic class. The central result is not a numerical mismatch for the special choice

$F(x) = 2\log x$. It is a no-regularity theorem for every function F in this final-denominator-only class.

Our contributions are four precise statements.

1. We separate oriented open double cosets from closed conjugacy classes and give exact group and even Gauss-word witnesses showing that $|c(g)|$ is not a closed-orbit invariant.

2. We prove that the square repetition law alone forces every $F : \alpha\mathbb{N}_{>0} \to \mathbb{R}$ to vanish. The proof uses a two-parameter positive hyperbolic family and assumes no continuity, monotonicity, or logarithmic form.

3. We compute the exact power defect of $2\log|c(g^n)|$ and prove that its stable homogenization is the Selberg translation length. This identifies the positive closure rather than merely rejecting the literal clock.

4. We show that an affine copy of Φ cannot become one entire Riemann ξ-function after multiplication by an entire zero-free normalizer. This is a supporting global-divisor obstruction, not the main theorem delta.

Most ingredients surrounding these statements are classical. Selberg zeta encodes closed hyperbolic classes [7]; Series relates modular geodesics to continued fractions [8]; Mayer realizes Selberg zeta through Gauss transfer operators [3, 4]; and modern cusp acceleration gives nuclear Fredholm realizations in broad generality [5]. Our contribution is a scoped compatibility theorem that prevents the classical scattering quotient from being promoted unchanged into a same-clock primitive-orbit or Hilbert–Pólya construction.

The paper first records the classical open-channel arithmetic, then imposes the axioms of a closed period. The resulting rigidity theorem is followed by the stable Selberg closure, the divisor assessment, and an independently checked computational ledger. Every universal claim is proved symbolically; finite computation is used only to audit implementation and conventions.

2 Open modular scattering arithmetic

Let $\Gamma = \mathrm{PSL}_2(\mathbb{Z})$ and let $P = \Gamma_\infty = \langle T\rangle$ be the cusp stabilizer, where

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Calculations use lifts $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ in $\mathrm{SL}_2(\mathbb{Z})$. In the big Bruhat cell we choose the PSL sign by requiring $c > 0$.

Proposition 2.1 (Oriented cusp double cosets)**.** *The cell with $c = 0$ is the identity double coset P. The nonidentity double cosets in $P\backslash\Gamma/P$ are in bijection with*

$$\{(c, \bar{d}) : c \geq 1,\ \bar{d} \in (\mathbb{Z}/c\mathbb{Z})^\times\}.$$

Consequently, the number of oriented algebraic channels at denominator c is $\varphi(c)$.

Proof. Left multiplication by T^m leaves the bottom row fixed, while right multiplication by T^n sends (c, d) to $(c, d + nc)$. Hence c and $d \bmod c$ are double-coset invariants. The determinant equation $ad - bc = 1$ implies $(c, d) = 1$. Conversely, for a unit $\bar{d} \bmod c$, choose a with $ad \equiv 1 \pmod{c}$ and put $b = (ad - 1)/c$.

For uniqueness, use a right translation to make two representatives with the same label have the same bottom row. The difference of their top rows is an integer multiple of that primitive bottom row, so a left translation makes the matrices equal. If $c = 0$, determinant one forces the diagonal entries to be both 1 or both -1, and the PSL class lies in P. ☐

Assign the open height

$$\tau_P(PgP) = 2\log c(g).$$

Then proposition 2.1 gives, initially for $\Re s > 1$,

$$\sum_{PgP\neq P} e^{-s\tau_P(PgP)} = \sum_{c\geq 1} \frac{\varphi(c)}{c^{2s}} = \frac{\zeta(2s-1)}{\zeta(2s)}. \tag{1}$$

The last equality follows from $\varphi(n) = n\sum_{d|n}\mu(d)/d$ and absolute rearrangement of the Dirichlet series. This arithmetic series is not the complete scattering coefficient. The constant term of the normalized Eisenstein series is

$$E(z,s) = y^s + \Phi(s)y^{1-s} + \text{nonconstant modes},$$

with

$$\Phi(s) = \sqrt{\pi}\frac{\Gamma(s-\frac{1}{2})}{\Gamma(s)}\frac{\zeta(2s-1)}{\zeta(2s)} = \frac{\Lambda(2s-1)}{\Lambda(2s)}. \tag{2}$$

The Eisenstein/zeta relation is classical [10].

Guillemin introduced sojourn times as the geometric clock of scattering geodesics [1], and Ji–Zworski related scattering matrices to scattering geodesics on locally symmetric spaces [2]. In the modular case, recent explicit counting uses a clock $2\log(cT_0)$ after a cusp cutoff T_0 [6]. This supports the denominator as an *open* clock. The count $\varphi(c)$ uses oriented algebraic double cosets; identifying a geometric trajectory with its reversal can change the multiplicity convention.

Nothing in this section defines a primitive closed orbit. In particular, Pg^nP is not declared to be an n-fold repeat of PgP. Powers enter only after one proposes to descend final-monodromy data to the closed category.

3 Why closure changes the indexing object

A closed geodesic on the modular surface is indexed here by a hyperbolic conjugacy class. We retain the orientation encoded by the conjugacy class and do not quotient $[g]$ by inversion $[g^{-1}]$; making the unoriented quotient would not affect the matrix obstruction below. A total period L must be representative independent and obey

$$L([g^n]) = nL([g]).$$

The two indexing categories impose different invariance conditions.

	Open scattering channel	Closed hyperbolic orbit
Index	PgP	conjugacy class $[g]$
Invariant data	$(c, d \bmod c)$	trace/eigenvalue and cyclic word
Clock	$2\log(cT_0)$	$2\,\mathrm{arcosh}(\lvert\operatorname{tr} g\rvert/2)$
Repetition	not imposed on $P\backslash\Gamma/P$	$[g^n]$ traverses $[g]$ n times

Proposition 3.1 (Failure of conjugacy invariance)**.** *The function $g \mapsto |c(g)|$ does not descend to hyperbolic conjugacy classes in* $\mathrm{PSL}_2(\mathbb{Z})$.

Proof. Take

$$g = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}, \qquad S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Then

$$S^{-1}gS = \begin{pmatrix} 3 & -2 \\ -1 & 1 \end{pmatrix}.$$

The two matrices have trace 4 and are conjugate in $\mathrm{PSL}_2(\mathbb{Z})$, but their absolute lower-left entries are 2 and 1. $\square$

The same failure appears inside the continued-fraction coding of closed geodesics. Let

$$A_a = \begin{pmatrix} 0 & 1 \\ 1 & a \end{pmatrix}.$$

The even words $(1,1,1,2)$ and $(1,2,1,1)$ give

$$A_1A_1A_1A_2 = \begin{pmatrix} 2 & 5 \\ 3 & 8 \end{pmatrix}, \qquad A_1A_2A_1A_1 = \begin{pmatrix} 3 & 5 \\ 4 & 7 \end{pmatrix}.$$

With $Q = A_1A_1 = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, the second product is Q^{-1} times the first times Q. The conjugator has determinant one, the traces agree, and the denominators are 3 and 4. The two-digit shift keeps the witness inside PSL rather than relying on a single-digit PGL conjugacy. Series' coding makes cyclic recoding the natural closed-geodesic equivalence [8].

Remark 3.2 (Scope of the type test). A local roof need not be a class function. Only its full periodic Birkhoff sum must descend to a closed orbit. Thus proposition 3.1 rules out a *total* period that depends only on the final denominator. It does not rule out local denominator increments, cyclic sums, coboundary corrections, or endpoint-extended state spaces.

4 Denominator-only repetition rigidity

Conjugacy failure already rejects the literal entry as a total period, but it does not exclude a function that might erase the representative dependence. Call a matrix *entry-positive* when all four entries are strictly positive. The next theorem closes the final-denominator-only class once the standard repetition axiom is imposed; it needs the square law only on the entry-positive subfamily.

Theorem 4.1 (Scaled denominator-only square rigidity). *Fix $\alpha > 0$ and let $F : \alpha\mathbb{N}_{>0} \to \mathbb{R}$ be arbitrary. Suppose*

$$F(\alpha|c(g^2)|) = 2F(\alpha|c(g)|) \tag{3}$$

for every entry-positive hyperbolic $g \in \mathrm{SL}_2(\mathbb{Z})$. Then

$$F(\alpha n) = 0 \qquad (n \geq 1).$$

Proof. For $m, n \geq 1$, set

$$\gamma_{m,n} = \begin{pmatrix} 1 & m \\ n & 1+mn \end{pmatrix}.$$

It has determinant one and trace $2 + mn > 2$. Direct multiplication gives

$$c(\gamma_{m,n}) = n, \qquad c(\gamma_{m,n}^2) = n(2+mn).$$

First take $n = 1$. Equation (3) yields

$$F(\alpha r) = 2F(\alpha) \qquad (r \geq 3). \tag{4}$$

Next take $(m,n) = (1,r)$ for any $r \geq 3$. The square law gives

$$F(\alpha r(r+2)) = 2F(\alpha r).$$

Both arguments in (4), apart from the common α, are at least 3. Substitution gives $2F(\alpha) = 4F(\alpha)$, hence $F(\alpha) = 0$. Equation (4) now gives $F(\alpha r) = 0$ for every $r \geq 3$.

Finally take $(m,n) = (1,2)$. Since $c(\gamma_{1,2}^2) = 8$, the square law reads $F(8\alpha) = 2F(2\alpha)$. Its left side is zero, so $F(2\alpha) = 0$. Together with $F(\alpha) = 0$, this covers all positive integers. □

The theorem requires neither continuity nor monotonicity and uses only squares. Full power homogeneity contains the square law, so we immediately obtain the intended Euler-product obstruction.

Corollary 4.2 (Closed Euler-product obstruction)**.** *There is no nontrivial primitive-hyperbolic Euler product whose total period has the form $R_F(g) = F(\alpha|c(g)|)$ for fixed $\alpha > 0$ and whose repeats satisfy $R_F(g^n) = nR_F(g)$ on every hyperbolic class.*

Proof. The repetition axiom implies (3); hence theorem 4.1 gives $F \equiv 0$. All proposed periods vanish and the associated local norm $e^{R_F(g)}$ is identically one, which is degenerate rather than a nontrivial standard dynamical Euler product. □

The corollary does not affect the open Dirichlet series (1), because that series is indexed by double cosets, not primitive hyperbolic classes. It also does not apply when the norm uses trace, eigenvalues, a complete word, endpoint data, or an enlarged cocycle.

5 Stable closure returns to Selberg length

The literal denominator height fails exact repetition, but its growth under powers has a canonical linear part. Cayley–Hamilton identifies it without a fit.

Proposition 5.1 (Chebyshev power identity)**.** *For $g \in \mathrm{SL}_2(\mathbb{Z})$, $t = \operatorname{tr} g$, and $n \geq 1$,*

$$g^n = U_{n-1}(t/2)g - U_{n-2}(t/2)I,$$

where $U_{-1} = 0$, $U_0 = 1$, and $U_{k+1}(x) = 2xU_k(x) - U_{k-1}(x)$. In particular,

$$c(g^n) = c(g)U_{n-1}(t/2). \tag{5}$$

Proof. The characteristic polynomial of g is $x^2 - tx + 1$, so $g^2 = tg - I$. The claimed formula holds at $n = 1, 2$. Multiplication by g and substitution of $g^2 = tg - I$ turns the step from n to $n + 1$ into the defining recurrence for U_n. Taking lower-left entries proves (5). □

Choose the lift of a hyperbolic class with $t > 2$, and define

$$\lambda = \frac{t + \sqrt{t^2 - 4}}{2}, \qquad \ell(g) = 2\log\lambda, \qquad H_\alpha(g) = 2\log(\alpha|c(g)|).$$

Hyperbolicity forces $c(g) \neq 0$.

Theorem 5.2 (Exact defect and stable homogenization)**.** *For every $n \geq 1$,*

$$H_\alpha(g^n) = n\ell(g) + 2\log\frac{\alpha|c(g)|}{\sqrt{t^2 - 4}} + 2\log(1 - \lambda^{-2n}). \tag{6}$$

Consequently,

$$\lim_{n\to\infty} \frac{H_\alpha(g^n)}{n} = \ell(g).$$

Proof. The eigenvalues are λ and λ^{-1}, and

$$U_{n-1}(t/2) = \frac{\lambda^n - \lambda^{-n}}{\lambda - \lambda^{-1}} = \frac{\lambda^n(1 - \lambda^{-2n})}{\sqrt{t^2 - 4}}.$$

Combine this identity with (5), multiply by α, and take $2\log$. This proves (6). After division by n, both defect terms tend to zero. □

The limiting clock is exactly the translation length used in the Selberg–Mayer closed-geodesic determinant [7, 4]. This positive closure explains why enforcing power stability removes the endpoint denominator information and returns to eigenvalue growth.

The conclusion is not a universal uniqueness theorem. It concerns the canonical limit of the specific sequence $H_\alpha(g^n)/n$. A useful scoped rigidity statement does follow: if a homogeneous closed clock L satisfies

$$L([g^n]) - H_\alpha(g^n) = o(n)$$

along the powers of a fixed g, then division by n and theorem 5.2 give $L([g]) = \ell(g)$. Homogeneous class functions outside this asymptotic class remain unrestricted.

6 Divisor obstruction and Route-A assessment

The clock theorem already stops the proposed closed Euler product. The scattering coefficient also has a global analytic obstruction to being one entire Riemann target under source-allowed normalizations.

Define

$$\Lambda(u) = \pi^{-u/2}\Gamma(u/2)\zeta(u), \qquad \xi(u) = \tfrac{1}{2}u(u-1)\Lambda(u).$$

Then

$$\Phi(s) = \frac{s}{s-1}\frac{\xi(2s-1)}{\xi(2s)}. \tag{7}$$

Theorem 6.1 (Zero-free normalization cannot remove the shifted divisor)**.** *Let $a, b \in \mathbb{C}$ with $a \neq 0$, and let h be entire and zero-free. Then $h(s)\Phi(as+b)$ is not entire. In particular, it cannot equal the entire function $\xi(s)$ as a global meromorphic identity.*

Proof. Let ρ be a nontrivial zero of ζ, of multiplicity m. At $s = \rho/2$, the denominator $\Lambda(2s)$ vanishes. The numerator is $\Lambda(\rho - 1) = \Lambda(2-\rho)$ by the functional equation. Since $\Re(2-\rho) > 1$, this value is nonzero. Thus Φ has a pole of multiplicity m at $\rho/2$.

At $s = (1+\rho)/2$, the numerator vanishes and the denominator $\Lambda(1+\rho)$ is nonzero, so Φ has a shifted zero. The affine map $s \mapsto as+b$ is bijective, and an entire zero-free h is finite and nonzero at every transported pole. Hence multiplication by h cannot remove the poles. □

A cusp-coordinate change $y' = ry$, followed by renormalization of the incoming coefficient to one, sends $\Phi(s)$ to $r^{2s-1}\Phi(s)$. The extra exponential is entire and zero-free, so it lies inside theorem 6.1. A compensator with its own zeros or a meromorphic zeta divisor could cancel poles, but would introduce arithmetic not supplied by a zero-free cusp normalization. The theorem is global and does not prohibit a local identity on a pole-free domain.

The resulting Route-A decision is deliberately negative for the candidate and positive for the obstruction package.

Gate	Verdict	Decisive reason
A1 primitive object	fail	open channel $\neq$ closed class
A2 same-clock determinant	fail	repetition forces $F = 0$
A3 global analytic target	fail	two shifted divisors
A4 operator lift	not testable	no same-clock operator

The physical-line scattering matrix is nevertheless a natural unitary operator-valued object. That classical structure does not turn its resonant divisor into a discrete self-adjoint Hilbert–Pólya point spectrum. Route B is therefore not invoked.

7 Exact audit, reproducibility, and limits

The computational layer is designed to catch convention and implementation errors, not to infer the universal theorems from a census. The producer uses exact integer matrix arithmetic for double cosets, conjugacy witnesses, the positive rigidity family, and Chebyshev identities. The library `mpmath` is used only for frozen high-precision evaluations of formulas proved above.

Audit block	Size	Result
Double-coset levels	80	all counts equal $\varphi(c)$
Positive rigidity family	400	all square identities exact
Chebyshev power rows	48	all identities exact
Primitive even Gauss words	274	259 cyclic variations
Literal denominator square law	274	0 passes
Stable-formula rows ($\alpha = 1$)	96	maximum residual $< 2.70 \cdot 10^{-79}$
Dirichlet control rows	12	all inside elementary tail bounds

At $s = 1/2 + it$ for $t = 0.7, 2, 7$, the producer checks $|\Phi(s)| = 1$ and $\Phi(s)\Phi(1-s) = 1$ with maximum reported residual $2.11 \cdot 10^{-81}$. These are normalization regressions, not evidence about unlisted zeta zeros.

A second checker does not import the producer. At 110 decimal digits it reimplements matrix multiplication, totients, double-coset representatives, Gauss-word enumeration, Chebyshev powers, translation length, the Dirichlet series, and the scattering coefficient. It verifies every row in the six producer artifacts. Regression tests also mutate a certificate and require the checker to reject it. No prime table, Riemann-zero table, fitted scale, or target-derived weight is read.

The numerical defect audit fixes $\alpha = 1$; general α adds the constant $2 \log \alpha$ in equation (6) and does not alter the analytic limit. The audit has three further limitations. First, the 274-word count is a finite illustration; theorem 4.1 carries the universal quantifier. Second, high-precision residuals check an implementation of equations (2) and (6); they do not prove those identities. Third, a self-issued artifact report is not a cryptographic signature. Git provenance and a release manifest provide the external reproducibility anchor.

The mathematical scope is equally narrow. The no-go theorem does not cover local denominator cocycles, cyclic sums, cohomological corrections, trace-dependent or complete-word clocks, endpoint-extended transfer operators, matrix cocycles, subadditive pressure, open groupoid traces, multi-cusp scattering, or divisor-carrying compensators. Each escape changes the source-locked object and requires a separate Route-A evaluation.

8 Conclusion

The modular cusp denominator is an exact arithmetic scattering coordinate, but it is the wrong type of datum for a final-monodromy closed period. The entry-positive hyperbolic family proves more than failure of the logarithmic ansatz: every denominator-only function satisfying the square repetition law is zero. Power stabilization does recover a meaningful closed clock, and the result is exactly Selberg translation length.

This combination closes a direct route from the classical scattering zeta quotient to a same-clock primitive Euler product or Hilbert–Pólya operator. It does not close cusp dynamics as a research direction. The next substantial tests should change the mathematical object: retain endpoints in an open groupoid, build a genuinely local chronological cocycle, or use a source-locked multi-cusp scattering matrix. Increasing the current word cutoff or fitting zeros would not address the proved obstruction.

A Reproducibility protocol

Run the following commands from the project directory:

```
python code/modular_clock.py --output results
python code/independent_check.py --results results \
  --output results/independent_check.json
python -m unittest discover -s code -p 'test_*.py' -v
```

The default source lock is: double-coset cutoff 80, rigidity-family cutoff 20 in each parameter, Gauss digits 1 through 3 and even word lengths through 6, powers through 24, Dirichlet cutoff 50,000, and producer precision 80 decimal digits. The independent checker uses 110 digits. The 274-word census discards proper powers, then retains the lexicographically least representative of each even-cyclic-rotation class. This even rotation convention keeps the associated conjugators in $\mathrm{SL}_2(\mathbb{Z})$. The stable-formula CSV fixes $\alpha = 1$.

The primary machine-readable files are:

```
exact_certificates.json
double_coset_counts.csv
gauss_word_clock_audit.csv
homogenization.csv
dirichlet_convergence.csv
summary.json
independent_check.json
```

Integer fields are exact. Decimal strings store high-precision evaluations; they are not rounded binary floats. The producer's elementary Dirichlet tail bound uses $\varphi(c) \leq c$, and the independent checker recomputes both the partial sums and the stated bound. The code reads no external data file.

References

[1] Victor W. Guillemin. Sojourn times and asymptotic properties of the scattering matrix. *Publications of the Research Institute for Mathematical Sciences*, 12(99):69–88, 1976.

[2] Lizhen Ji and Maciej Zworski. Scattering matrices and scattering geodesics of locally symmetric spaces. *Annales Scientifiques de l'École Normale Supérieure*, 34(3):441–469, 2001.

[3] Dieter H. Mayer. On the thermodynamic formalism for the gauss map. *Communications in Mathematical Physics*, 130(2):311–333, 1990.

[4] Dieter H. Mayer. The thermodynamic formalism approach to selberg's zeta function for PSL(2,z). *Bulletin of the American Mathematical Society*, 25(1):55–60, 1991.

[5] Anke D. Pohl and Paul Wabnitz. *Selberg Zeta Functions, Cuspidal Accelerations, and Existence of Strict Transfer Operator Approaches*, volume 318 of *Memoirs of the American Mathematical Society*. American Mathematical Society, 2026. Preprint arXiv:2209.05927.

[6] Sudhir Pujahari and Punya Plaban Satpathy. Prime scattering geodesic theorem, 2025. arXiv:2505.04973.

[7] Atle Selberg. Harmonic analysis and discontinuous groups in weakly symmetric riemannian spaces with applications to dirichlet series. *Journal of the Indian Mathematical Society*, 20:47–87, 1956.

[8] Caroline Series. The modular surface and continued fractions. *Journal of the London Mathematical Society*, 31(1):69–80, 1985.

[9] Liang Wang. An area-preserving hénon-map model for the riemann zeros: A deterministic-dynamics approach with quantum and dissipative solvers. Foundational manuscript in the local project corpus; repository `https://github.com/maris205/hilbert-polya-structure`, file 5-An Area-Preserving Henon-Map Model.pdf, accessed 2026-08-07, 2026.

[10] Don Zagier. Eisenstein series and the riemann zeta function. In *Automorphic Forms, Representation Theory and Arithmetic*, pages 275–301. Tata Institute of Fundamental Research, Bombay, 1981.